Introduction

The hazard function $h(t)$ gives the instantaneous rate at which events occur at time t, given that they have not occurred before t. It is the primary object in survival analysis and reliability engineering: proportional-hazards models, accelerated-failure-time models, and competing-risks methods all reason about hazards rather than distributions directly.

Prerequisites

Continuous distributions; survival function $S(t) = P(T > t)$.

Theory

For a non-negative continuous random variable T with density f and survival function $S(t) = 1 - F(t)$,

$$h(t) = \frac{f(t)}{S(t)}.$$

Intuitively, $h(t)\,dt$ is approximately $P(t \le T < t + dt \mid T \ge t)$ for small dt.

The **cumulative hazard** is

$$H(t) = \int_0^t h(u)\,du = -\log S(t).$$

Conversely, $S(t) = e^{-H(t)}$. Once either of f, F, S, h, H is known, the others follow.

Hazard shapes for common distributions:

- Exponential: $h(t) = \lambda$ (constant).
- Weibull with shape k: $h(t) = (k/\lambda)(t/\lambda)^{k-1}$ – increasing if $k > 1$, decreasing if $k < 1$.
- Gompertz: $h(t) = \alpha e^{\beta t}$ – exponentially increasing; common in human mortality.
- Log-normal: $h(t)$ rises then falls.

Assumptions

T is a non-negative continuous random variable with a density. Proportional-hazards models further assume $h(t \mid x) = h_0(t)\exp(\beta^\top x)$.

R Implementation

```r
library(survival); library(ggplot2)

# Constant hazard (exponential)
x <- seq(0.01, 5, length.out = 400)
h_exp <- rep(0.5, length(x))
h_weibull_inc <- (2.0 / 1.5) * (x / 1.5)^(2.0 - 1)
h_weibull_dec <- (0.5 / 1.5) * (x / 1.5)^(0.5 - 1)

df <- data.frame(
  t  = rep(x, 3),
  h  = c(h_exp, h_weibull_inc, h_weibull_dec),
  kind = factor(rep(c("exp lambda=0.5",
                      "weibull k=2 (incr)",
                      "weibull k=0.5 (decr)"), each = length(x)))
```

```r
)

ggplot(df, aes(t, h, colour = kind)) +
  geom_line(linewidth = 1) +
  scale_colour_manual(values = c("#6A4C93", "#2A9D8F", "#F4A261")) +
  labs(x = "t", y = "h(t)", colour = "",
       title = "Hazard functions for three canonical models") +
  theme_minimal()

# Empirical cumulative hazard from censored data
set.seed(2026)
T <- rexp(100, rate = 0.3)
status <- rep(1, 100)
fit <- survfit(Surv(T, status) ~ 1)
plot(fit, fun = "cumhaz", col = "#2A9D8F", lwd = 2,
     xlab = "t", ylab = "H(t)",
     main = "Empirical cumulative hazard (Nelson-Aalen)")
abline(0, 0.3, col = "#F4A261", lty = 2)
```

Output & Results

The plot shows three hazards: exponential (flat), increasing Weibull, decreasing Weibull. The empirical cumulative hazard from 100 exponential event times is approximately linear with slope $\lambda = 0.3$, matching the theoretical $H(t) = 0.3t$.

Interpretation

In survival analysis, hazards are the natural target of regression: "the hazard of death was 40 % higher in the treated group than in controls (HR = 1.40, 95 % CI 1.10-1.78)". The proportional-hazards model is ubiquitous because it avoids specifying the baseline hazard shape.

Practical Tips

- A constant hazard implies exponentiality; a log-log plot of $-\log S(t)$ against $\log t$ that is linear indicates Weibull.
- Empirical hazards are noisy at the tail where few subjects remain; report cumulative hazard (smoother) or kernel-smoothed hazard estimates.
- The **Nelson-Aalen** estimator is the standard non-parametric estimator of the cumulative hazard under right-censoring.
- In competing risks, cause-specific hazards differ from the sub-distribution hazard of Fine-Gray; the two answer different questions.
- Time-varying hazard ratios invalidate the proportional-hazards assumption; use stratification or time-interactions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots